

Lignan contents of Dutch plant foods: a database including lariciresinol, pinoresinol, secoisolariciresinol and matairesinol.



Enterolignans (enterodiol and enterolactone) can potentially reduce the risk of certain cancers and cardiovascular diseases. Enterolignans are formed by the intestinal microflora after the consumption of plant lignans. Until recently, only secoisolariciresinol and matairesinol were considered enterolignan precursors, but now several new precursors have been identified, of which lariciresinol and pinoresinol have a high degree of conversion. Quantitative data on the contents in foods of these new enterolignan precursors are not available. Thus, the aim of this study was to compile a lignan database including all four major enterolignan precursors. Liquid chromatography-tandem mass spectrometry was used to quantify lariciresinol, pinoresinol, secoisolariciresinol and matairesinol in eighty-three solid foods and twenty-six beverages commonly consumed in The Netherlands. The richest source of lignans was flaxseed (301,129 microg/100 g), which contained mainly secoisolariciresinol. Also, lignan concentrations in sesame seeds (29,331 microg/100 g, mainly pinoresinol and lariciresinol) were relatively high. For grain products, which are known to be important sources of lignan, lignan concentrations ranged from 7 to 764 microg/100 g. However, many vegetables and fruits had similar concentrations, because of the contribution of lariciresinol and pinoresinol. Brassica vegetables contained unexpectedly high levels of lignans (185-2321 microg/100 g), mainly pinoresinol and lariciresinol. Lignan levels in beverages varied from 0 (cola) to 91 microg/100 ml (red wine). Only four of the 109 foods did not contain a measurable amount of lignans, and in most cases the amount of lariciresinol and pinoresinol was larger than that of secoisolariciresinol and matairesinol. Thus, available databases largely underestimate the amount of enterolignan precursors in foods.